

PSG COLLEGE OF TECHNOLOGY
DEPARTMENT OF COMPUTER APPLICATIONS
23MX26 - Java Programming Laboratory
Hands-on Worksheet 7

Problem 1 — Uniform Store Shirt Management

Your college uniform store sells shirts of different sizes.

Each shirt has:

- collar size
- sleeve length
- material (default cotton)

The store manager wants a small Java program to create shirt objects and display their details.

Requirements from Problem Statement:

Create a class Shirt with:

- int collarSize
 - int sleeveLength
 - String material = "cotton"
1. Constructor to initialize collarSize and sleeveLength
 2. Method displayDetails()
 3. Create 3 Shirt objects in TestShirt
 4. Display all details

Sample Output

Shirt 1 → Collar: 38, Sleeve: 30, Material: cotton

Shirt 2 → Collar: 40, Sleeve: 32, Material: cotton

Shirt 3 → Collar: 42, Sleeve: 34, Material: cotton

Skeleton Code

```
class Shirt {
    int collarSize;
    int sleeveLength;
    String material = "cotton";

    Shirt(int c, int s) {
        // initialize
    }
    void displayDetails() {
        // print values
    }
}

public class TestShirt {
    public static void main(String[] args) {
        ----
    }
}
```

Viva Questions

1. What is the purpose of constructor?
2. Where is material initialized?
3. How many objects are created here?

Problem 2 — Mobile Phone Inventory

An electronics shop stores details of mobile phones in stock.

Each phone has:

- brand

- model
- price

The owner wants to calculate discount price.

Problem Statement

Create Mobile class:

- Constructor initializes all fields
- Methods:
 - display()
 - getDiscountPrice(int percent)

Create 3 mobiles and show discounted prices.

Sample Output

Samsung M14 Price: 15000

After 10% discount: 13500

Skeleton Code

```
class Mobile {
    String brand;
    String model;
    double price;

    Mobile(String b, String m, double p) {

    }

    void display() {

    }

    double getDiscountPrice(int percent) {
        return 0;
    }
}
```

Viva Questions

1. Why return type is double?
 2. Does method modify original price?
 3. Difference between local and instance variables?
-

Problem 3 — Library Book Tracker

A small library wants to track books borrowed by students.

Each book has:

- title
- author
- availability status

Students can borrow and return books.

Problem Statement

Create Book class with:

- Constructor
- Methods:
 - borrowBook()
 - returnBook()

- `displayStatus()`

Sample Output

Book borrowed successfully
Status: Not Available
Book returned
Status: Available

Skeleton Code

```
class Book {  
    String title;  
    String author;  
    boolean available;  
  
    Book(String t, String a) {  
        ....  
    }  
  
    void borrowBook() {  
        ....  
    }  
  
    void returnBook() {  
        ....  
    }  
  
    void displayStatus() {  
  
    }  
}
```

Viva Questions

1. Why use boolean for availability?
 2. What changes when book is borrowed?
 3. Is each book an independent object?
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Problem 4 — Employee Salary Calculator (Slightly Higher Thinking)

A company stores employee details and calculates monthly salary.
Each employee has:

- id
- name
- basic salary

Salary = basic + allowance (10%)

Problem Statement

Create Employee class:

- Constructor to initialize values
- Methods:
 - `calculateSalary()`
 - `displayDetails()`

Create 3 employees and print salary.

Sample Output

Skeleton Code

```
class Employee {  
    int id;  
    String name;  
    double basic;  
  
    Employee(int i, String n, double b) {  
  
    }  
  
    double calculateSalary() {  
        return 0;  
    }  
  
    void displayDetails() {  
  
    }  
}
```

Viva Questions

1. Why calculation inside method?
2. What is object state here?
3. Can two employees share same data?